

## CSC10006 – Introduction to Database

# Advanced Nested Query

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# 1 Objectives

After completing this practical lesson, students will know:

- How to represent division operation in relational algebra using SQL
- How to represent set operations (subtraction, intersection, union) using SQL

## 2 Detailed Instructions

### 2.1 Set Operations

#### 2.1.1 Subtraction

Methods:

- Use EXCEPT operator
- Nested query with NOT EXISTS or NOT IN

**Example 2: Find teachers who do not participate in any project.**

Observation:

Teachers not participating in projects = All teachers – Teachers participating in projects.

Using EXCEPT:

```
-- Simple: Only shows MAGV information
SELECT MAGV
FROM GIAOVIEN GV
EXCEPT
SELECT MAGV
FROM THAMGIADT TG

-- Complex: Shows MAGV and HOTEN information
SELECT MAGV, HOTEN
FROM GIAOVIEN GV
EXCEPT
SELECT GV.MAGV, GV.HOTEN
FROM THAMGIADT TG, GIAOVIEN GV
WHERE TG.MAGV = GV.MAGV
```

Using NOT EXISTS:

```
SELECT GV.MAGV, GV.HOTEN
FROM GIAOVIEN GV
WHERE NOT EXISTS ( SELECT *
                    FROM THAMGIADT TG
                    WHERE TG.MAGV = GV.MAGV)
```

Using NOT IN:

```
SELECT MAGV, HOTEN
FROM GIAOVIEN
WHERE MAGV NOT IN (SELECT MAGV
                   FROM THAMGIADT)
```

### 2.1.2 Intersection

Methods:

- Use INTERSECT operator
- Nested query with EXISTS or IN
- Use regular join operation

**Example 3: Find teachers who both participate in projects and are department heads.**

Observation: Result = Teachers participating in projects  $\cap$  Teachers who are department heads.

Using INTERSECT:

```
SELECT TRUONGBM FROM BOMON
INTERSECT
SELECT MAGV FROM THAMGIADT
```

Using ... IN (...) AND ... IN (...):

```
SELECT GV.MAGV, GV.HOTEN
FROM GIAOVIEN GV
WHERE MAGV IN (SELECT TRUONGBM FROM BOMON) AND
      MAGV IN (SELECT MAGV FROM THAMGIADT)
```

Using EXISTS (...) AND EXISTS (...):

```

SELECT GV.MAGV, GV.HOTEN
FROM GIAOVIEN GV
WHERE EXISTS (SELECT * FROM BOMON WHERE TRUONGBM=GV.MAGV) AND
EXISTS (SELECT * FROM THAMGIADT TG WHERE TG.MAGV = GV.MAGV)

```

Using regular join:

```

SELECT TG.MAGV
FROM BOMON BM, THAMGIADT TG
WHERE BM.TRUONGBM = TG.MAGV

```

### 2.1.3 Union

Methods:

- Use UNION operator (duplicate rows will be removed) / UNION ALL (takes all rows from tables)
- Nested query with EXISTS or IN

Conditions for performing UNION / INTERSECT,  
EXCEPT: Tables must have the same number of attributes  
and corresponding data types between columns.

The output table columns are the columns from the first table.

| <u>TABLE1</u> | <u>TABLE2</u> |
|---------------|---------------|
| a             | b             |
| 1             | 2             |
| 3             | 4             |
| 2             | 3             |
| 4             | 5             |
| a             | b             |
| 2             | 7             |
| 3             | 4             |
| 2             | 3             |
| 1             | 6             |

| UNION                                                                                                                                                                                                                                        | UNION ALL                                                         |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SELECT * FROM TABLE1<br><br>UNION<br><br>SELECT * FROM TABLE2                                                                                                                                                                                | SELECT * FROM TABLE1<br><br>UNION ALL<br><br>SELECT * FROM TABLE2 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Kết quả : <table> <tr><th>a</th><th>b</th></tr> <tr><td>1</td><td>2</td></tr> <tr><td>1</td><td>6</td></tr> <tr><td>2</td><td>3</td></tr> <tr><td>2</td><td>7</td></tr> <tr><td>3</td><td>4</td></tr> <tr><td>4</td><td>5</td></tr> </table> | a                                                                 | b | 1 | 2 | 1 | 6 | 2 | 3 | 2 | 7 | 3 | 4 | 4 | 5 | Kết quả : <table> <tr><th>a</th><th>b</th></tr> <tr><td>1</td><td>2</td></tr> <tr><td>3</td><td>4</td></tr> <tr><td>2</td><td>3</td></tr> <tr><td>4</td><td>5</td></tr> <tr><td>2</td><td>7</td></tr> <tr><td>3</td><td>4</td></tr> <tr><td>2</td><td>3</td></tr> <tr><td>1</td><td>6</td></tr> </table> | a | b | 1 | 2 | 3 | 4 | 2 | 3 | 4 | 5 | 2 | 7 | 3 | 4 | 2 | 3 | 1 | 6 |
| a                                                                                                                                                                                                                                            | b                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                            | 2                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                            | 6                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2                                                                                                                                                                                                                                            | 3                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2                                                                                                                                                                                                                                            | 7                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                                                                                                                            | 4                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4                                                                                                                                                                                                                                            | 5                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| a                                                                                                                                                                                                                                            | b                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                            | 2                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                                                                                                                            | 4                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2                                                                                                                                                                                                                                            | 3                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 4                                                                                                                                                                                                                                            | 5                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2                                                                                                                                                                                                                                            | 7                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 3                                                                                                                                                                                                                                            | 4                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 2                                                                                                                                                                                                                                            | 3                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1                                                                                                                                                                                                                                            | 6                                                                 |   |   |   |   |   |   |   |   |   |   |   |   |   |                                                                                                                                                                                                                                                                                                          |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

**Example 4: List teachers who participate in projects or teachers who are department heads.**

Using UNION:

```
SELECT MAGV FROM THAMGIADT
UNION
SELECT TRUONGBM FROM BOMON
```

Using EXISTS (...) OR EXISTS (...):

```
SELECT GV.MAGV, GV.HOTEN
FROM GIAOVIEN GV
WHERE EXISTS ( SELECT * FROM BOMON WHERE TRUONGBM=GV.MAGV) OR
      EXISTS ( SELECT * FROM THAMGIADT TG WHERE TG.MAGV = GV.MAGV)
```

Using IN (...) OR IN (...):

```
SELECT GV.MAGV, GV.HOTEN
FROM GIAOVIEN GV
WHERE GV.MAGV IN (SELECT TRUONGBM FROM BOMON ) OR
      GV.MAGV IN ( SELECT MAGV FROM THAMGIADT TG )
```

## 2.2 Division Operation

Given tables R (A, B, C, D, E) and S (D, E) as follows:

| R | A        | B | C        | D | E |
|---|----------|---|----------|---|---|
|   | $\alpha$ | a | $\alpha$ | a | 1 |
|   | $\alpha$ | a | $\gamma$ | a | 1 |
|   | $\alpha$ | a | $\gamma$ | b | 1 |
|   | $\beta$  | a | $\gamma$ | a | 1 |
|   | $\beta$  | a | $\gamma$ | b | 3 |
|   | $\gamma$ | a | $\gamma$ | a | 1 |
|   | $\gamma$ | a | $\gamma$ | b | 1 |
|   | $\gamma$ | a | $\beta$  | b | 1 |

| S     | D | E |
|-------|---|---|
| $b_i$ | a | 1 |
|       | b | 1 |

Dividend: R

Divisor: S

| R+S   | A        | B | C        |
|-------|----------|---|----------|
| $a_i$ | $\alpha$ | a | $\gamma$ |
|       | $\gamma$ | a | $\gamma$ |

### 2.2.1 Using NOT EXISTS ... EXCEPT

Formula:

```
SELECT R1.A, R1.B, R1.C
FROM R R1
WHERE NOT EXISTS (
    ( SELECT S.D, S.E FROM S)
    EXCEPT
    ( SELECT R2.D, R2.E
      FROM R R2
      WHERE R1.A=R2.A AND R1.B=R2.B AND R1.C=R2.C )
)
```

**Example 5: Find teachers (MAGV) who participate in all projects**

Dividend: THAMGIADT (MAGV, MADT, ...)

Divisor: DETAI (MADT, ...)

```
SELECT DISTINCT TG1.MAGV
FROM THAMGIADT TG1
WHERE NOT EXISTS (
    ( SELECT MADT FROM DETAI)
    EXCEPT
    ( SELECT MADT
      FROM THAMGIADT TG2
      WHERE TG2.MAGV = TG1.MAGV )
)
```

### 2.2.2 Using NOT EXISTS ... NOT EXISTS

Formula:

```
SELECT R1.A, R1.B, R1.C
FROM R R1
WHERE NOT EXISTS (
    SELECT *
    FROM S
    WHERE NOT EXISTS (
        SELECT *
```

```

FROM R R2
WHERE R2.D=S.D AND R2.E=S.E AND
      R1.A=R2.A AND R1.B=R2.B AND R1.C=R2.C

```

```
)
```

```
)
```

### Example 6: Find teachers (MAGV) who participate in all projects (Using NOT EXISTS)

Dividend: THAMGIADT (MAGV, MADT, ...)

Divisor: DETAI (MADT, ...)

```

SELECT TG.MAGV
FROM THAMGIADT TG1
WHERE NOT EXISTS (
    SELECT *
    FROM DETAI DT
    WHERE NOT EXISTS (
        SELECT *
        FROM THAMGIADT TG2
        WHERE TG2.MADT=DT.MADT AND TG2.MAGV=TG1.MAGV
    )
)

```

### 2.2.3 Using GROUP BY with COUNT

For R(A,B), S(B), performing  $R \div S$

```

SELECT R.A
FROM R
[WHERE R.B IN (SELECT S.B FROM S [WHERE <CONDITION>])]
GROUP BY R.A
HAVING COUNT(DISTINCT R.B) = ( SELECT COUNT(S.B)
                                FROM S
                                [WHERE <CONDITION>])

```

### Example 7: Find teachers (MAGV) who participate in all projects (Using GROUP BY)

Dividend: THAMGIADT (MAGV, MADT, STT, ...)

Divisor: DETAI (MADT, ...)



```

SELECT TG.MAGV
FROM THAMGIADT TG1
GROUP BY TG.MAGV
HAVING COUNT(DISTINCT TG1.MADT) = ( SELECT COUNT(DT.MADT)
                                   FROM DETAI DT )

```

## 2.3 Additional Examples

**Example 9: Find names of 'HTTT' teachers who participate in all projects with topic 'QLGD'**

```

-- EXCEPT
SELECT DISTINCT TG1.MAGV, GV.HOTEN
FROM THAMGIADT TG1, GIAOVIEN GV
WHERE TG1.MAGV=GV.MAGV AND GV.MABM='HTTT'
      AND NOT EXISTS (
                                ( SELECT MADT FROM DETAI WHERE MACD='QLGD')
                                EXCEPT
                                ( SELECT MADT
                                  FROM THAMGIADT TG2
                                  WHERE TG2.MAGV = TG1.MAGV )
                        )

-- NOT EXISTS
SELECT TG1.MAGV, GV.HOTEN
FROM THAMGIADT TG1, GIAOVIEN GV
WHERE TG1.MAGV=GV.MAGV AND GV.MABM='HTTT'
      AND NOT EXISTS ( SELECT *
                                FROM DETAI DT
                                WHERE MACD='QLGD'
                                AND NOT EXISTS ( SELECT *
                                                  FROM THAMGIADT TG2
                                                  WHERE TG2.MADT=DT.MADT AND
                                                    TG2.MAGV=TG1.MAGV
                                                )
                        )

```

```
-- COUNT
SELECT TG1.MAGV , GV.HOTEN
FROM THAMGIADT TG1 , GIAOVIEN GV
WHERE TG1.MAGV=GV.MAGV AND GV.MABM='HTTT'
      AND TG1.MADT IN (SELECT MADT FROM DETAI WHERE MACD='QLGD')
GROUP BY TG1.MAGV , GV.HOTEN
HAVING COUNT(DISTINCT TG1.MADT) = ( SELECT COUNT(MADT)
                                   FROM DETAI
                                   WHERE MACD='QLGD' )
```

### 3 In-Class Exercises

Requirements: Write queries Q58, Q60, ..., Q72 in the ‘Project Management’ exercise. Each query should be written in 3 ways: Using EXCEPT, NOT EXISTS, and COUNT.

Duration: 02 hours.

- Q58. Find the names of teachers who participate in projects covering all topics.
- Q59. Find the names of projects in which all teachers from the HTTT department participate.
- Q60. Find the names of projects in which all teachers from the "Hệ thống thông tin" department participate.
- Q61. Find teachers who have participated in all projects with topic code QLGD.
- Q62. Find the names of teachers who participate in all projects that teacher “Trần Trà Hương” has participated in.
- Q63. Find the names of projects in which all teachers from the “Hoá hữu cơ” department participate.
- Q64. Find the names of teachers who participate in all tasks of project 006.
- Q65. Find teachers who have participated in all projects of the “Ứng dụng công nghệ” topic.
- Q66. Find the names of teachers who have participated in all projects lead by teacher “Trần Trà Hương”.
- Q67. Find the names of projects in which all teachers from the “CNTT” faculty participate.
- Q68. Find the names of teachers who participate in all tasks of the “Nghiên cứu tế bào gốc” project.
- Q69. Find the names of teachers assigned to all projects with budgets over 100 million?
- Q70. Find the names of projects in which all teachers from the “Sinh học” faculty participate.

- Q71. Find the ID, name, and date of birth of teachers who participate in all tasks of the "Ứng dụng hoá học xanh" project.
- Q72. Find the ID, name, department name, and academic supervisor name of teachers who participate in all projects under the "Nghiên cứu phát triển" topic.